

# Landscape effects on large scale abundance patterns of turtle doves *Streptopelia turtur* in Portugal

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**Abstract** The European turtle dove is both a highly valued game species and a species of conservation concern, which is declining due probably to a combination of habitat degradation and unsustainable hunting. Although declines seem to be less severe in the Mediterranean region, it remains uncertain the extent to which ongoing land use changes will negatively affect this species. This study examined this issue, by estimating the effects of landscape composition on the broad scale abundance pattern of breeding turtle doves in continental Portugal. Turtle doves were surveyed in the breeding seasons of 2002 and 2003, from 3160 point counts spaced at about 1-km intervals along 158 transects of about 20 km, evenly covering the country. The frequency of occurrence of turtle doves at each transect was used as a proxy of species abundance, and related using GAM modelling to 21 variables describing land cover and woody linear features (e.g., hedgerows and riparian galleries). Turtle doves were most abundant in north- and central-

eastern Portugal, with high abundances also recorded in the regions around Lisbon and along the Guadiana valley. Abundances were positively related to forest cover, particularly by broadleaved forests and by pine stands without woody understory, to cover by permanent crops, and to the density of woody linear habitats. Results suggest that conservation of Mediterranean turtle doves requires policies and management strategies reversing the pervasive trends of forest management neglect and agricultural abandonment, while preserving hedgerows and riparian galleries in more intensive agricultural landscapes.

**Keywords** Conservation · Forest management · Habitat selection · GAM · Mediterranean landscapes

## Introduction

Some bird species are both legally hunted and have unfavourable conservation status, making its management particularly challenging (Bro et al. 2001; Machado et al. 2008; Besnard et al. 2010). This is the case of the European turtle dove (*Streptopelia turtur*), a sub-Saharan migrant bird that has markedly declined throughout much of its European breeding range during the past decades (PECBMS 2009), and which has thus been considered a Species of European Conservation Concern (SPEC) (Jarry 1994). Reasons for the decline are still not completely understood, but they may include the degradation of breeding and wintering habitats, along with unsustainable hunting pressure (Browne and Aebischer 2005; Lutz and Jensen 2007; Eraud et al. 2009). Thus, understanding the factors driving the species occurrence and abundance remains a prerequisite for proper population conservation and management (Lutz and Jensen 2007). This is particularly important in Euro-Mediterranean countries, where the population is still large but may be at risk from rapid changes in land cover

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and agricultural practices (Moreira and Russo 2007; Stoate et al. 2009; Bugalho et al. 2011).

In North and Central Europe, the turtle dove is mostly associated with agricultural landscapes, where it nests in scrublands, woodland edges, hedgerows and thickets (Hermant and Frochot 1997; Aubineau and Boutin 1998; Rouxel 2000; Browne and Aebischer 2005). Agriculture intensification has been considered one of the main causes for the observed reductions in turtle dove's breeding performance and population declines in this region, probably due to the loss of adequate nesting and foraging habitats (Dunn and Morris 2012). In contrast, the contribution of land use changes to turtle dove's population declines in the Euro-Mediterranean region is uncertain, because of insufficient information on the species' habitat requirements. The few studies available suggested that the species is affected by broad land cover patterns (Seoane et al. 2004), occurring in both forest and agricultural landscapes (Fontoura and Dias 1996; Hermant and Frochot 1997; Bakaloudis et al. 2009). Within forest landscapes, turtle doves appear to be influenced by both forest type (e.g., broadleaved versus conifers) and management, often showing higher densities in forests with little understory cover (Rocha and Hidalgo 2002; Martinez 2004; Camprodon and Brotons 2006; Bakaloudis et al. 2009). In agricultural landscapes of the Mediterranean region, turtle doves occupy a wide variety of habitats, reaching particularly high densities in permanent crops such as olive or orange groves, and other dry and irrigated orchards (Peiró 1990; Boukhemza-Zemmouri et al. 2008; Hanane and Baamal 2011). Agricultural management seems also to influence turtle dove densities, as some specific crop types appear to be preferred for feeding or nesting (Dias and Fontoura 1996; Rocha and Hidalgo 2002). Despite these data, most studies in the Mediterranean have been carried out at small scales, and thus their results may reflect local or regional ecological specificities. It thus remains uncertain whether the documented habitat preferences can also affect abundance patterns at large spatial scales.

The present study examined the influence of landscape composition on the abundance patterns of the turtle dove across continental Portugal, which holds ca. 10 % of the European breeding population and accounts for ca. 6–12 % of the annual bagged turtle doves in Europe (Lutz and Jensen 2007). Although the distribution of this species in Portugal was documented by the recent national breeding bird atlas (Equipa Atlas 2008), no study has hitherto detailed the variation in abundance patterns and habitat associations across the country. This information would be important for the conservation management of the species, given the strong to moderate population declines registered in the last decades (Catre et al. 2010), and the potentially detrimental habitat trends associated with land abandonment, agricultural intensification, and urbanization (Freire et al. 2009; Feranec et al. 2010; Bugalho et al. 2011). The study was

based on a detailed national survey specifically directed at breeding turtle doves, aiming at (1) characterising the broad scale abundance patterns across the country, and (2) examining the importance of landscape features in explaining the observed abundance patterns. Results were then used to discuss landscape management strategies favouring the conservation of turtle doves in Portugal and elsewhere.

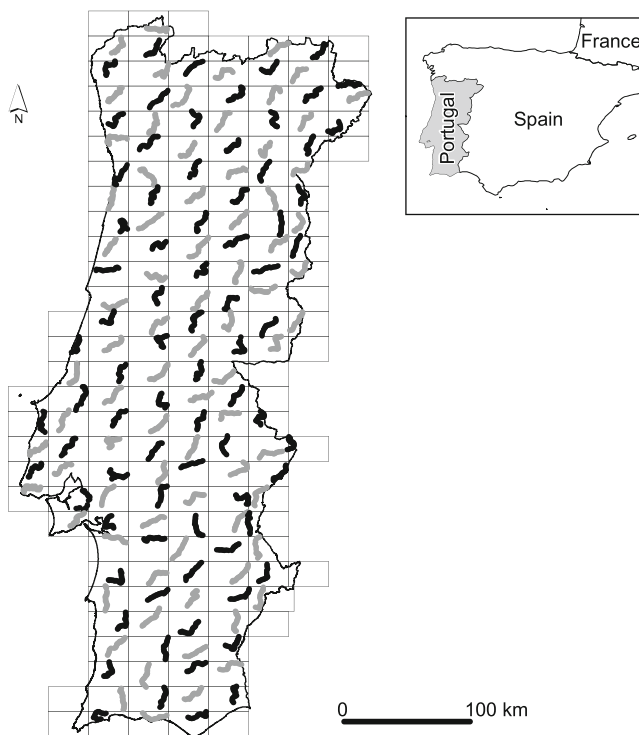
## Methods

### Study area

The study was conducted in continental Portugal, which includes a variety of climates, relief, vegetation types and land use patterns despite its rather small size (ca. 80,000 km<sup>2</sup>). Although the climate and vegetation are predominantly Mediterranean, there is marked spatial variation in environmental conditions across the country, with a wide range of altitudes (sea level to 1,993 m), total annual precipitation (400–2,800 mm), and mean annual temperature (7–18°C). Land cover includes mostly farmland (33 %), forests (39 %) and shrubland (21 %) (Direcção Nacional de Gestão Florestal 2010). Forested areas are primarily dominated by plantations of maritime pine *Pinus pinaster* (27 %) and eucalyptus *Eucalyptus* spp. (23 %), which are most widespread in the northern half of the country, and native forests of cork oak *Quercus suber* (23 %) and holm oak *Quercus rotundifolia* (13 %), which are more widespread in the south and are often part of agro-silvo-pastoral systems (Direcção Nacional de Gestão Florestal 2010). Farmland is dominated by pastures (48 %), arable crops (32 %), and permanent crops (19 %) such as olive groves, vineyards and fruit orchards (INE 2010). Human population and urban areas are largely concentrated along the coast (Freire et al. 2009). There is high variation in landscape composition and configuration, resulting from a complex interplay between the abiotic environment, disturbance–succession vegetation dynamics, and land use history spanning several millennia. Further information on the study area can be consulted in APA (2012).

### Turtle dove surveys

Abundance patterns of the turtle dove were estimated in continental Portugal using point counts set along road transects (point-transect technique; Bibby et al. 2000), following an approach comparable to that used in other large scale bird surveys (e.g., Sauer et al. 2011). To obtain a spatially homogeneous distribution of sampling effort, the country was divided in 158 rectangular 20×32 km cells, and one transect was sampled per grid cell (Fig. 1). Due to logistical constraints, half of the cells were surveyed in 2002 and the remaining in 2003, following a checkerboard pattern in each



**Fig. 1** Transects sampled during the springs of 2002 (black lines) and 2003 (grey lines) to estimate turtle dove abundance patterns in continental Portugal

year. Transects were about 20 km (mean  $\pm$  standard error:  $22.0 \pm 2.3$  km) long, and they were demarcated before surveys along secondary roads using 1:25,000 military maps, aiming to obtain a route of sufficient length that (1) was as linear and continuous as possible, (2) was oriented from northeast to southeast, (3) crossed rural areas, and (4) was minimally interrupted by urban areas. These criteria were established so that transects were as similar as possible across grid cells, thereby enhancing comparability and reducing potential sources of random variation.

In each transect, we counted turtle doves at 20 sampling points. The first point coincided with the beginning of the transect; all the other points were set 1 km apart of each other. This spacing was considered sufficient to avoid double counts (e.g., Boutin et al. 2001). Points could be in forest, shrubland or farmland habitats, but urban areas and major water bodies (e.g., large rivers and reservoirs) were excluded. When points fell in one of these habitats, they were shifted 250 m away from their limits, though maintaining a minimum distance of 1 km between points. At each sampling point, turtle dove vocalisations and sightings were recorded for 5 min within 250 m of the observer (Bibby et al. 2000). To avoid potential biases due to changes in bird detectability across the breeding season, counts were concentrated in May, based on preliminary observations suggesting that most turtle doves have already arrived and started establishing breeding territories by the beginning of this month (Fontoura and Dias 1996). In

addition, counts were usually carried out from sunrise to 10 AM during the morning peak of bird activity, thereby avoiding time-of-day effects (Calladine et al. 1999). Finally, counts were not carried out during adverse weather conditions such as rainfall or strong wind (Bibby et al. 2000).

Estimates of turtle dove abundance were made at the transect scale, by combining data from all point counts within each transect. The proportion of point counts where turtle doves were recorded (hereafter referred to as prevalence) was used as an abundance index instead of the total number of turtle dove contacts per transect, because the two metrics were highly correlated (Spearman's  $R=0.95$ ,  $p<0.001$ ), and the former was considered the most robust as it is less prone to inter-observer variability.

### Landscape composition

Factors influencing turtle dove abundance patterns were explored using a set of variables reflecting landscape composition in terms of dominant land cover categories (Table 1). Particular care was taken to quantify the presence or absence of woody understory in forested habitats, and the density of woody linear habitats, as these were shown in previous studies to strongly influence turtle dove distribution and abundance (e.g., Aubineau and Boutin 1998; Browne and Aebischer 2005; Bakaloudis et al. 2009). The proportion of each land cover category was estimated during turtle dove surveys, within 250-m radius circles centred on each point count. Each circle was divided into eight sectors (octants), and the dominant land cover in each sector was visually estimated. The 160 octant evaluations per transect were then used to estimate the proportion of each land cover category in each transect. The density of linear woody habitats in each transect was indexed from the proportion of sampling points where tree lines, hedgerows and riparian gallery occurred.

### Data analysis

The patterns of turtle dove abundance across continental Portugal were estimated by spatial interpolation, following an approach widely used to produce robust maps of relative species abundances from bird counts along transects (Bahn and McGill 2007; Sauer et al. 2011). Before interpolation, data from the two annual surveys were pooled, and the data of a given transect was attributed to its geographical centre. Interpolation was carried out using kriging (e.g., Goovaerts 1997; Webster and Olivier 2001), which is considered particularly adequate to estimate a spatial pattern from sample locations (Maurer 1994). The procedure started by calculating the sample variogram describing the variability in turtle dove's prevalence between pairs of points separated by increasingly larger distances. A spherical semivariogram

**Table 1** Description and summary statistics of variables used to analyse landscape factors affecting turtle dove abundance in continental Portugal

| Variable                  | Description                                                                     | Mean | SD   | Range  |
|---------------------------|---------------------------------------------------------------------------------|------|------|--------|
| Agricultural habitats     | Crops and pastures                                                              | 0.51 | 0.31 | 0–1.00 |
| Grassland                 | Pastures, fallows and stubbles                                                  | 0.15 | 0.14 | 0–0.66 |
| Annual crops              | Cereals, vegetables and other arable crops                                      | 0.08 | 0.10 | 0–0.58 |
| Permanent crops           | Vineyards, olive groves and other orchards                                      | 0.13 | 0.17 | 0–0.91 |
| Oak savannahs             | Cork and holm oak parklands with shrubs, pastures or annual crops as understory | 0.14 | 0.22 | 0–0.81 |
| Forest habitats           | All forest types                                                                | 0.19 | 0.25 | 0–0.90 |
| Forests with scrub        | Forests with woody understory                                                   | 0.15 | 0.19 | 0–0.88 |
| Forest without scrub      | Forest without woody understory                                                 | 0.14 | 0.16 | 0–0.72 |
| Broadleaved               | Broadleaved forests (excluding eucalyptus)                                      | 0.05 | 0.09 | 0–0.50 |
| Broadleaves with scrub    | Broadleaved forests with woody understory                                       | 0.03 | 0.06 | 0–0.46 |
| Broadleaves without scrub | Broadleaved forests without woody understory                                    | 0.02 | 0.04 | 0–0.29 |
| Eucalyptus                | Eucalyptus plantations                                                          | 0.07 | 0.12 | 0–0.84 |
| Eucalyptus with scrub     | Eucalyptus plantations with woody understory                                    | 0.03 | 0.06 | 0–0.46 |
| Eucalyptus without scrub  | Eucalyptus plantations without woody understory                                 | 0.04 | 0.09 | 0–0.52 |
| Conifers                  | Conifer forests, mainly pine plantations                                        | 0.12 | 0.16 | 0–0.79 |
| Conifers with scrub       | Conifer forests with woody understory                                           | 0.07 | 0.12 | 0–0.78 |
| Conifers without scrub    | Conifer forests with woody understory                                           | 0.06 | 0.10 | 0–0.54 |
| Mixed                     | Forests with both conifers and broadleaves                                      | 0.05 | 0.08 | 0–0.37 |
| Mixed with scrub          | Mixed forest with woody understory                                              | 0.03 | 0.06 | 0–0.36 |
| Mixed without scrub       | Mixed forest without woody understory                                           | 0.02 | 0.04 | 0–0.26 |
| Scrublands                | Tall and dwarf shrublands                                                       | 0.10 | 0.13 | 0–0.64 |
| Woody linear habitats     | Density index of tree lines, hedgerows and riparian galleries                   | 0.17 | 0.21 | 0–0.90 |

Variables are the proportions of each land cover category estimated in 158 transects sampled for turtle dove abundance, except where indicated otherwise

was used to represent the sample variogram, as it is thought to mimic species dispersal more realistically than other theoretical functions (Goovaerts 1997). Different values for the semivariogram parameters (lag, sill and range) were tested until they accounted for the maximum amount of variation in the raw data (Webster and Olivier 2001). Cross-validation was used to check model performance, by kriging at each sampling point in turn after excluding the sample value there (Goovaerts 1997; Webster and Olivier 2001). The scatter of the true values against their estimates was then plotted, and a simple linear regression model was fit between the two variables, with the coefficient of determination used to indicate the proportion of variation in the data accounted for by the model. Although the regression model is expected to be statistically significant, the coefficient of determination may be relatively low, because kriging provides a smoothed surface, underestimating the larger values and overestimating the smaller ones (Webster and Olivier 2001). The model selected was the one that best attained the following goals simultaneously (ESRI 2005): (1) the mean prediction error should be close to 1; (2) the average standard error and the root-mean-squared (RMS)

prediction error should be similar and to be as small as possible and (3) the root-mean squared standardized error (RMSS) should be close to one. Interpolation was based on the nearest five neighbours within the interpolation window. Analyses were carried out using the Geostatistical Analyst extension in ESRI's ArcGIS®9.1 (ESRI 2005).

Prior to the analysis of factors influencing the turtle dove's abundance patterns, the arcsine transformation was used on all landscape variables to approach normality and to reduce the influence of a few large values. A principal component analysis (PCA) of land cover variables was then used to investigate multicollinearity and to describe dominant landscape gradients (Legendre and Legendre 1998). A varimax normalized rotation was applied to the set of principal components (PC) with eigenvalues >1, to obtain simpler and more interpretable ecological gradients (Legendre and Legendre 1998). The site coordinates along the PCs were mapped to visualise gradients of landscape composition across the country. The variable describing the density of woody linear structures was excluded from PCA, in order to analyse its effects separately from those of the dominant land cover categories.

Relationships between turtle dove prevalence and landscape variables were explored using Generalised Additive Models (GAM), thereby accounting for potential nonlinearities in species responses (Wood 2006; Zuur et al. 2009). The analysis was based on the regression on principal components approach (Legendre and Legendre 1998), using the axis extracted from the landscape PCA, the density of woody linear habitats and the sampling year as explanatory variables. All PCs with eigenvalues  $>1$  were used in the analysis to explore the eventual effects of the main landscape composition gradients, but in the final model only those most related to the independent variable were retained (Legendre and Legendre 1998; see below). Year was included to control for eventual changes in overall turtle dove prevalence between sampling years. Pairwise correlations between each PC axis and each of the other two variables were all  $\leq |0.30|$ , indicating that there were no collinearity problems. As the response variable was a proportion, we used a binomial GAM with a logit link. The dispersion parameter ( $\phi$ ) of the model including all explanatory variables (full model) was larger than 1 ( $\phi = 4.17$ ), so standard errors were corrected for overdispersion using a quasi-binomial GAM.

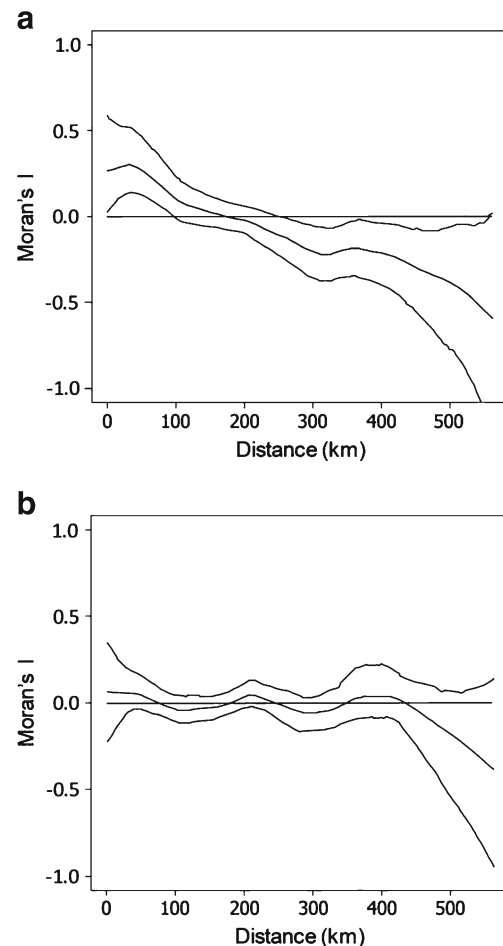
The modelling procedure involved the fitting of the full model, followed by backward elimination of non-significant ( $p > 0.05$ ) variables (Zuur et al. 2009). Model adequacy was evaluated by plotting residuals versus fitted values and explanatory variables. Model fit was evaluated by the proportion of the null deviance explained. The *mgcv* package (Wood 2006) for R 2.14.1 (R Development Core Team 2011) was used to fit GAM using penalised regression splines with the optimal amount of smoothing estimated by generalised cross validation (GCV). The significance of spatial autocorrelation in the patterns of turtle dove's abundance and in GAM scaled Pearson residuals was tested using spline correlograms based on Moran's I (Bjornstad and Falck 2001) with 95 % confidence intervals, estimated by the *ncf* package (Bjornstad 2009) for R.

## Results

### Turtle dove distribution patterns

Turtle doves occurred in 66.5 % of 158 sampling transects, with an average prevalence per transect of  $17.0 \pm 1.6$  % (range 0–80 %). Using only data from transects where the species occurred, the average prevalence was  $26.0 \pm 1.8$  %. There was significant positive spatial autocorrelation in turtle dove prevalence up to a distance of ca. 100 km, and a negative autocorrelation above distances of ca. 250 km (Fig. 2a).

The kriged surface depicting spatial patterns in turtle dove abundance was generated using an isotropic spherical semivariogram model with nugget effect (see Table S1 in Online Resource 1 for details). The kriged surface was



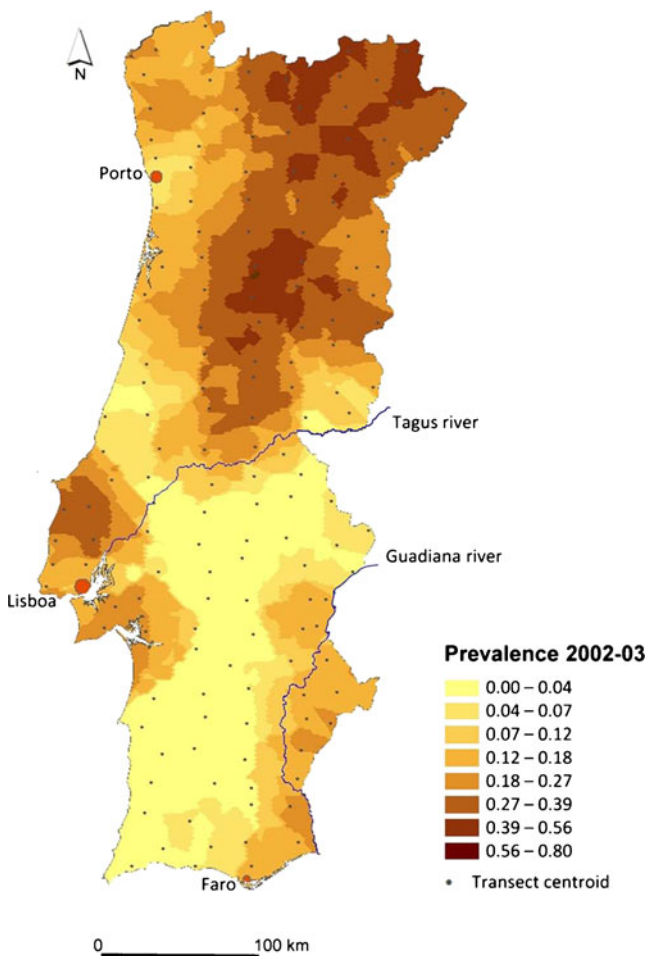
**Fig. 2** Spline correlograms (Moran's  $I$ ), with 95 % pointwise bootstrap confidence intervals, of turtle dove prevalence in continental Portugal (a), and of scaled Pearson residuals of the GAM model (Table 3) relating landscape variables to turtle dove prevalence (b)

considered satisfactory, as indicated by the kriging error coefficients (Table S1 in Online Resource 1). In cross-validation there was a significant linear relationship between observed and predicted values ( $F_{1, 156} = 39.172$ ,  $p < 0.001$ ), though the determination coefficient was relatively low because of the smoothing effect of kriging ( $r^2 = 0.201$ ). The interpolated map indicated that turtle doves were most common in the north- and central-eastern parts of the country, with prevalence decreasing further south (Fig. 3). However, there were a few spots with higher prevalence in the coastal areas around Lisbon, as well as along the Guadiana river valley, in the southeast (Fig. 3).

### Overall landscape patterns

The landscape PCA summarised the 21 original variables into six axes (PC) with eigenvalues  $> 1$ , together accounting for 82.1 % of variance in the data (Table 2), and reflecting clear spatial gradients in landscape composition across the





**Fig. 3** Kriged surface depicting spatial variation in breeding turtle dove prevalence in continental Portugal, built from data collected at 158 transects sampled in 2002 and 2003 (see Online Resource 1). Prevalence classes assigned according to eight quantiles from light yellow (low prevalence) to dark brown (high prevalence)

country (Fig. S1 in Online Resource 2). The first PC (hereafter  $PC_{\text{Forests with scrub}}$ ) depicted a gradient of increasing cover by forests with well-developed woody understory, particularly conifer forests, which were most common in the northern and central parts of the country. The second PC ( $PC_{\text{Forests no scrub}}$ ) represented cover by forests without woody understorey, also dominated by conifer forests and showing a spatial pattern similar to  $PC_{\text{Forests with scrub}}$ . The third PC expressed the cover by broadleaved forests ( $PC_{\text{Broadleaves}}$ ), which were most common in the northeast of the country. The fourth and fifth PC axes described gradients of increasing cover by eucalyptus plantations ( $PC_{\text{Eucalyptus}}$ ) and mixed forests ( $PC_{\text{Mixed}}$ ), respectively. The former were most prevalent along the coast, mainly in the northern coastal strip, whereas the later were most common near the coast, and in the region around Lisbon. The sixth PC represented a decreasing proportion of permanent crops ( $PC_{\text{Permanent}}$ ), which were most common in the south

coast, the Guadiana valley, the region north of Lisbon and also the northeast. Woody linear habitats structures were most prevalent in the northern border of the country, as well as in specific locations south of the Tagus river (Fig. S1 in Online Resource 2).

#### Landscape effects on turtle doves

The multivariate GAM of the turtle dove prevalence explained 30.4 % of the total deviance, and included the positive effect of forests without understory ( $PC_{\text{Forests no scrub}}$ ), broadleaved forests ( $PC_{\text{Broadleaves}}$ ) and linear woody habitats, which were stronger for the former two variables (Fig. 4a, b, c). There was also a strongly nonlinear effect (5.8 estimated degrees of freedom) of permanent crops ( $PC_{\text{Permanent}}$ ), suggesting a sudden increase in turtle dove abundance in landscapes with more than 40 % of this cover type (Fig. 4d). However, this section of the response curve was supported by few transects (Fig. 4d). There was no significant spatial autocorrelation in the residuals of the GAM model (Fig. 2b, Table 3), and inspection of diagnostic plots of residuals did not point to any significant problem.

#### Discussion

##### Turtle dove abundance patterns

This study showed that turtle doves are unevenly distributed in continental Portugal, with the highest abundances in the north- and central-eastern regions, and smaller abundance peaks in the areas around Lisbon and along the Guadiana valley in the south-east. These patterns are largely similar to those obtained in the recent Atlas of breeding birds in Portugal (Equipa Atlas 2008), which was based on a far more exhaustive spatial cover, though not specifically directed at any particular species. The matching of ours and the Atlas results suggest that the simple sampling protocol used herein, associated with spatial interpolation using kriging, offers a consistent picture of the turtle dove abundance patterns and core breeding areas, thus providing a basis for the regular monitoring of this declining species. Specifically, we suggest that a monitoring scheme for turtle doves in Portugal and eventually elsewhere, could be based on a network of permanent transects about 20 km long, where the prevalence of turtle doves should be estimated from point counts spaced at about 1-km intervals, sampled on a single occasion during the first 2–3 h after sunrise. The best sampling period should be the subject of further research, because recent data suggested that peak vocal activity occurs from mid-May to mid-June (S. Dias, unpublished data), though May might be favoured to allow consistency and thus comparability with the baseline established by this study (Calladine et al. 1999). Prevalence may be

**Table 2** Factor loadings and eigenvalues after a varimax normalized rotation, of axes with eigenvalues >1.0 extracted by a PCA of landscape variables, and the proportion of variance accounted for by each axis

| Variable                  | PC <sub>Forests with scrub</sub> | PC <sub>Forests no scrub</sub> | PC <sub>Broadleaves</sub> | PC <sub>Eucalyptus</sub> | PC <sub>Mixed</sub> | PC <sub>Permanent</sub> |
|---------------------------|----------------------------------|--------------------------------|---------------------------|--------------------------|---------------------|-------------------------|
| Agriculture               | −0.58                            | −0.58                          | −0.37                     | −0.23                    | −0.19               | −0.12                   |
| Grassland                 | −0.52                            | −0.35                          | 0.08                      | −0.31                    | −0.25               | 0.30                    |
| Annual crops              | −0.49                            | 0.05                           | −0.05                     | 0.04                     | −0.05               | 0.04                    |
| Permanent crops           | −0.13                            | −0.17                          | −0.17                     | −0.11                    | −0.11               | <b>−0.91</b>            |
| Oak savannas              | −0.28                            | −0.54                          | −0.45                     | −0.04                    | −0.01               | 0.33                    |
| Forest                    | 0.56                             | 0.53                           | 0.29                      | 0.40                     | 0.30                | 0.19                    |
| Forest with scrub         | <b>0.88</b>                      | 0.07                           | 0.32                      | 0.18                     | 0.16                | 0.12                    |
| Forest without scrub      | −0.12                            | <b>0.78</b>                    | 0.17                      | 0.43                     | 0.30                | 0.15                    |
| Conifers                  | 0.59                             | <b>0.76</b>                    | 0.04                      | 0.00                     | 0.09                | 0.13                    |
| Conifers with scrub       | <b>0.84</b>                      | 0.33                           | 0.14                      | −0.04                    | 0.02                | 0.13                    |
| Conifers without scrub    | 0.00                             | <b>0.90</b>                    | −0.08                     | 0.06                     | 0.09                | 0.06                    |
| Broadleaves               | 0.15                             | −0.01                          | <b>0.96</b>               | −0.07                    | 0.09                | 0.07                    |
| Broadleaves with scrub    | 0.27                             | −0.15                          | <b>0.85</b>               | −0.08                    | 0.05                | 0.06                    |
| Broadleaves without scrub | −0.05                            | 0.17                           | <b>0.83</b>               | −0.07                    | 0.08                | 0.06                    |
| Eucalyptus                | 0.09                             | 0.13                           | −0.14                     | <b>0.97</b>              | 0.02                | 0.04                    |
| Eucalyptus with scrub     | 0.46                             | −0.16                          | −0.08                     | <b>0.71</b>              | −0.05               | −0.06                   |
| Eucalyptus without scrub  | −0.19                            | 0.30                           | −0.14                     | <b>0.83</b>              | 0.03                | 0.11                    |
| Mixed                     | 0.36                             | 0.11                           | 0.05                      | 0.04                     | <b>0.90</b>         | 0.06                    |
| Mixed with scrub          | 0.60                             | −0.17                          | 0.03                      | 0.07                     | <b>0.65</b>         | 0.05                    |
| Mixed without scrub       | −0.09                            | 0.39                           | 0.06                      | −0.04                    | <b>0.81</b>         | 0.04                    |
| Scrublands                | 0.12                             | 0.39                           | 0.55                      | −0.22                    | −0.28               | 0.06                    |
| Eigenvalues               | 3.96                             | 3.69                           | 3.31                      | 2.76                     | 2.31                | 1.19                    |
| % variance                | 18.8                             | 17.6                           | 15.6                      | 13.1                     | 11.0                | 5.7                     |

Values in bold indicate factor loadings  $\geq |0.65|$

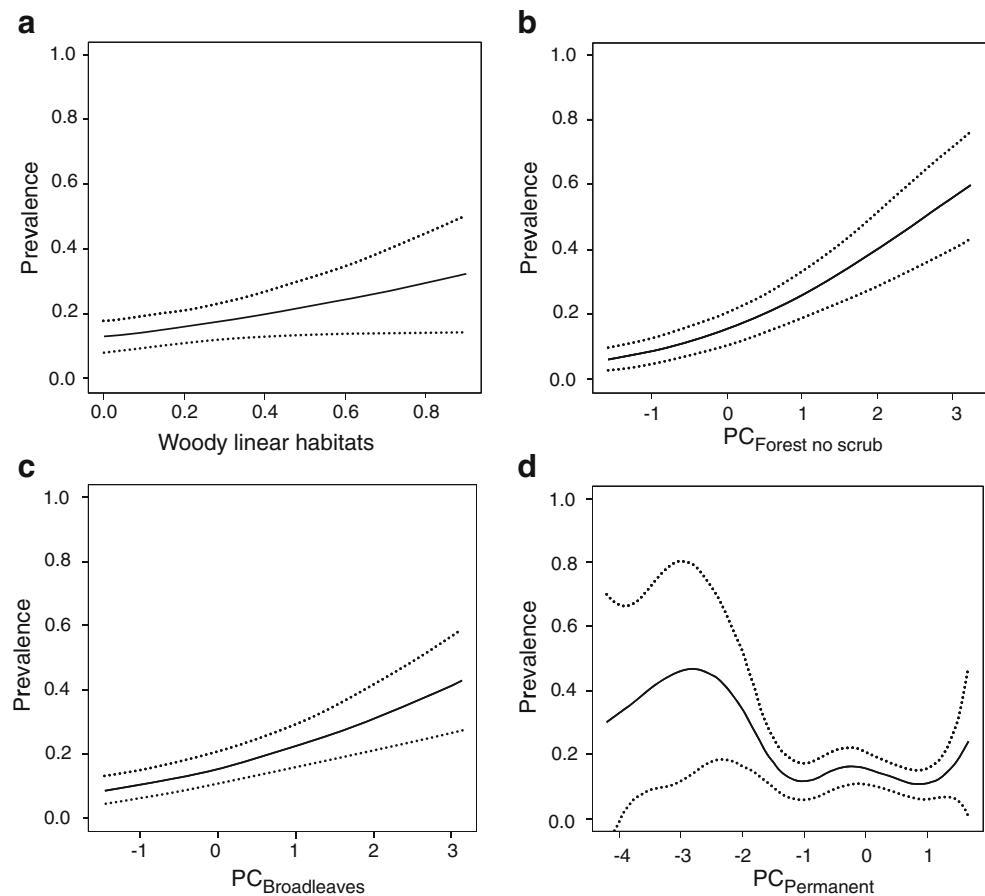
used as an index of breeding population size instead of actual turtle dove counts, as the two metrics were found to be highly correlated, and the former is likely to be less prone to observer biases. Implementing this management scheme would contribute to fulfil one of the requirements of the turtle dove management plan at European level (Lutz and Jensen 2007). Comparable cost-effective methods have been successfully used in dove monitoring programs, including, for instance, North America (Mourning dove call-count survey; Dolton and Rau 2005) and France (Boutin et al. 2001). More generally, transect data and spatial interpolation have been successfully used in other large scale and long term studies such as the North American Breeding Bird Survey (Sauer et al. 2011), providing a solid basis to generate maps of relative species abundances (Bahn and McGill 2007).

#### Landscape effects on turtle doves

Turtle doves in Portugal appeared to be strongly influenced by landscape composition, with the highest abundances associated with conifer plantations without woody understory, broadleaved forests, and permanent crops, and with high density of woody linear habitats. The positive association to these land

cover categories provided information on the types of land uses and landscape configurations that are favoured by the turtle dove. Conifer forests corresponded primarily to vast areas of maritime pine plantations in Northern and Central Portugal, where the absence of woody understory mostly results from management treatments to reduce fire hazard (Fernandes and Rigolot 2007). Broadleaved forests were primarily stands dominated by *Quercus* species, which usually are strongly fragmented and embedded in complex mosaic landscapes with scrubland and pastures (e.g., Moreira et al. 2001). Permanent crops were mostly olive groves and other orchards, which typically have an herbaceous understory. Woody linear habitats were mostly tree lines, hedgerows and riparian galleries, which are often associated with farmland landscapes where they are the main arboreal habitats. The study thus highlighted the preference of turtle doves for forests and agricultural landscapes with trees, including for instance orchards and arable land with riparian galleries and arboreal hedgerows, which is in line with previous research in Portugal (see Catry et al. 2010 for a review) and elsewhere (Rouxel 2000; Browne and Aebischer 2003; Martinez 2004; Bakaloudis et al. 2009; Buruaga et al. 2012; Hanane 2012). However, results from this and other studies contrast to some extent with the conventional view that the

**Fig. 4** Predicted values and 95 % CI (solid and dotted lines, respectively) of turtle dove prevalence in continental Portugal as a function of the variables entering in the final GAM: **a** Woody linear habitats; **b**  $PC_{\text{Forest no scrub}}$ ; **c**  $PC_{\text{broadleaves}}$ ; and **d**  $PC_{\text{Permanent}}$ . To estimate the curve for each variable the other variables were kept constant at their mean values



turtle dove is mainly a farmland species (BirdLife International 2004), suggesting instead that in the Mediterranean region the species may thrive over a wide range of habitat types (Peiró 1990; Rocha and Hidalgo 2002; Camprodon and Brotons 2006; Bakaloudis et al. 2009; Hanane and Baamal 2011; Buruaga et al. 2012; Hanane 2012).

The association of turtle doves with arboreal habitats was probably driven to a large extent by their breeding requirements, as the species mostly uses trees and tall shrubs for nesting (Nankinov 1994; Browne and Aebischer 2004; Camprodon and Brotons 2006; Bakaloudis et al. 2009;

Hanane 2012). However, habitats such as eucalyptus plantations, conifer plantations with woody understory, and mixed forests did not show any positive effect, suggesting that additional factors influenced the use of arboreal habitats by the turtle dove. A key factor was probably the availability of feeding resources close to nesting areas, as turtle doves are known to require mosaic landscapes where they can breed in arboreal habitats while foraging in nearby crops and grasslands (Browne and Aebischer 2004; Barbaro et al. 2007). This probably justifies the strong positive association of turtle doves with conifer plantations without woody understory, as these habitats can probably provide a suitable combination of secure nesting sites and food resources in the herbaceous understory. Seemingly, the positive responses to permanent crops could be due to the suitable nesting conditions in fruit trees, together with the favourable feeding conditions available in the herbaceous understory layer (Hanane and Baamal 2011; Hanane 2012). Understory effects could not be demonstrated for broadleaved forests, however, because the high correlation between cover by stands with and without woody understory made it impossible to disentangle the unique effects of each of these forest types. The avoidance of forest stands with dense woody understory by turtle doves was already reported in other studies carried out in the Iberian Peninsula (Santos et al.

**Table 3** Model relating turtle dove prevalence in continental Portugal to landscape variables (GAM with a logit link function)

| Parametric coefficients (intercept)  | Estimate<br>−1.76±0.11 | <i>t</i> value | <i>p</i> value<br><0.0001 |
|--------------------------------------|------------------------|----------------|---------------------------|
| Smooth terms                         | edf                    | F              | <i>p</i> value            |
| s (woody linear habitats)            | 1.000                  | 5.88           | 0.016                     |
| s ( $PC_{\text{Forests no scrub}}$ ) | 1.000                  | 32.53          | <0.0001                   |
| s ( $PC_{\text{Broadleaves}}$ )      | 1.000                  | 19.70          | <0.0001                   |
| s ( $PC_{\text{Permanent}}$ )        | 5.826                  | 2.29           | 0.039                     |

Deviance explained = 30.4 %, GVC score = 4.45

edf = estimated degrees of freedom



2002; Camprodon and Brotons 2006; Buruaga et al. 2012) and in Greece (Bakaloudis et al. 2009).

In contrast to forested landscapes, one of the key limiting factors for turtle doves in open farmland landscapes is probably the availability of suitable nesting sites (Aubineau and Boutin 1998; Browne and Aebischer 2005). In fact, turtle doves in these landscapes seem to be highly dependent on the presence of hedgerows and riparian galleries for nesting, while feeding in adjacent farmland habitats (Browne and Aebischer 2003, 2005; Dunn and Morris 2012). These observations possibly justify the strongly positive effects of woody linear habitats on the abundance of turtle doves in Portugal. Unlike northern Europe, where turtle doves often breed in tall and thorny bushes (e.g., hawthorn *Crataegus monogyna*) (Browne and Aebischer 2003; Dunn and Morris 2012), scrub may generally provide inadequate nest support in Mediterranean farmland, because the stature of shrubs dominating secondary shrublands (e.g., *Cistus*, *Ulex*, *Lavandula*, *Genista*, *Retama*) tends to be low and is thus unsuitable for most forest birds (Rey-Benayas et al. 2010).

Despite the value of the modelling exercise carried out in this study, the best GAM model explained only a relatively small part (ca. 30 %) of spatial variation in the abundance patterns of the turtle dove in Portugal. To account for this unexplained variation, future studies should include other important factors such as for instance climate variables which are known to play a key role in bird species distributions (e.g., Triviño et al. 2011). Also, modelling should consider fragmentation metrics in addition to the landscape composition variables used in this study, as there is increasing evidence that habitat fragmentation may play an important role in determining large scale species distributions (Reino et al. 2013). Finally, distribution models might be improved by including local features related for instance to microclimate, woody vegetation structure, or the availability and distribution of feeding habitats, which were unaccounted for in our study (Browne and Aebischer 2003, 2004; Bakaloudis et al. 2009). Overall, further research should thus combine a wider range of factors operating at several spatial scales, thereby gaining a better understanding on the causality of turtle dove abundance patterns.

### Conservation implications

In line with previous studies (Bakaloudis et al. 2009; Buruaga et al. 2012), our findings suggest that one of the critical habitats for Mediterranean turtle doves are conifer plantations with low cover by understory shrubs, which presently are at risk from the increasing frequency, extent and intensity of wildfires in the Mediterranean region (Moreira et al. 2011). These habitats are also threatened by management neglect, which results in rapid development of a dense woody understory layer (Santana et al. 2011),

strongly increasing fire hazard (Moreira et al. 2011) and making forests largely unsuitable for turtle doves and other ground-feeding and granivore bird species (Camprodon and Brotons 2006; Santana et al. 2012). In these circumstances, silvicultural treatments aimed at reducing fire hazard may be positive for turtle doves, as they often involve the removal of shrubs and small trees from the forest understory (Santana et al. 2011), thus promoting the development of the herbaceous layer (Porto et al. 2011). These fuel reduction treatments may have positive side effects for a wide range of organisms, though care should be taken to maintain untreated patches to protect late-successional forest species (Porto et al. 2011; Santana et al. 2011, 2012; Verdasca et al. 2012).

Large-scale shrub encroachment resulting from land abandonment seems also to be a problem in Portugal and elsewhere in the Mediterranean region, particularly in landscapes formerly devoted to extensive agriculture and agroforestry (Moreira and Russo 2007; Stoate et al. 2009; Bugalho et al. 2011), where it leads to the loss of landscape heterogeneity and declines in grassland habitats required by turtle doves for feeding. In Portugal, this process affects, for instance, a range of extensive olive groves and other permanent crops, thereby reducing the quality of a habitat strongly favoured by turtle doves. Land abandonment and subsequent scrub encroachment also promotes the occurrence of large wildfires, thereby reducing forest cover and thus the turtle dove nesting sites (Silva et al. 2011). Preventing or even reverting land abandonment trends is complex, but the problem may be minimised at least locally through financial incentives such as the agri-environment schemes, which support the maintenance of extensive agricultural practices (Stoate et al. 2009). The requirements of the turtle dove should thus be duly considered when designing and implementing agri-environment schemes.

In contrast to land abandonment, agriculture intensification promotes landscape homogenization through the removal of non-agricultural habitats such as hedgerows and riparian galleries, which are critical for the turtle dove persistence in farmland landscapes (Aubineau and Boutin 1998; Browne and Aebischer 2005; Buruaga et al. 2012). Regulatory frameworks may be required to halt this loss of woody linear habitats, while financial incentives such as the agri-environment schemes could contribute for recreating these features in intensively used landscapes where they have been lost in the past (Stoate et al. 2009). Besides favouring turtle dove habitats, these prescriptions could be positive for a wide range of farmland birds (Hinsley and Bellamy 2000), though care should be taken to avoid impacts on some farmland bird species of conservation concern which have a strongly negative response to woody edges (Reino et al. 2009).

In general, our results support the view that the turtle dove populations may occur over a wide range of forest and

agricultural landscapes. However, their persistence is conditional on the maintenance of either grassland patches or herbaceous understory layers in forested landscapes, and on the maintenance of tree lines, hedgerows or riparian galleries in otherwise open farmland. The species thus seems to be highly vulnerable to the ongoing dual trends of land abandonment and intensification that prevail in the Mediterranean region, thereby requiring policies and regulations promoting heterogeneity in forest and farmland landscapes. Besides protecting the turtle dove, these policies and regulations would have wider benefits for Mediterranean biodiversity (Moreira and Russo 2007; Stoate et al. 2009).

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